
List of Names:

- TVS, topological vector space;
- LCTVS, Local Complact Topological Space;

Chapter 1 Topological Vector Space

Preliminaries:

- General Topology;
- Filter;
- Linear Space, Linear Maps;

1.1 Definition

1.1.1 Linear Structure, and Compatibilities

Definition 1.1.1.1 Let E be a vector space over the field of complex field F . Let

$$\begin{aligned} A_v : E \times E &\rightarrow E & (x, y) &\mapsto x + y \\ M_s : F \times E &\rightarrow E & (\lambda, x) &\mapsto \lambda x \end{aligned}$$

Then, a topology τ on E is said to be **compatible with linear structure of E** if:

- $A_v : (E \times E, \tau \times \tau) \rightarrow (E, \tau)$ is continuous under product topology;
- $M_s : (F \times E, \mathcal{F} \times \tau) \rightarrow (E, \tau)$ is continuous under product topology.

Then,

Definition 1.1.1.2 A vector space compatible with linear structure of E is called **topological vector space(TVS)**.

Remark The topology is 'translation invariant'.

1.1.2 title

Definition 1.1.2.1

- (Absorbing)
- (Balanced)

► **Theorem** A filter $\mathcal{F}$ on the vector space E is the filter of neighborhoods of the origin in topology compatible with the linear structure if and only if it has the following properties:

- The origin belongs to every subset U belonging to $\mathcal{F}$;
- To every $U \in \mathcal{F}$
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◦ **Proof ...**

□

1.2 Hausdorff Topological Vector Spaces

Definition 1.2.0.1 ...

1.3 Quotient Topological Vector Spaces

1.4 Cauchy Filter, Complete Subsets, Completion

1.5 Compact Sets

Chapter 2 Semi-Norms, Norms

2.1 Semi-Norms

2.1.1 Convexity

Definition 2.1.1.1 ...

2.1.2 Barrel TVS

Definition 2.1.2.1 ...

2.1.3 LCTVS

Local Convexity

Definition 2.1.3.1 A TVS is said to be locally convex space if there is a basis of neighborhoods in E consisting of convex sets.

♠ **Proposition 2.1.3.2** ...

◦ **Proof** ...

□

2.2 Metrizable TVS

2.3 Norms

Chapter 3 Frechet Space

3.1 Definitions

3.1.1 F-Space

Definition 3.1.1.1 A **Frechet space** is a TVS with following properties:

- It is metrizable;
- Complete;
- Locally convex;

Sometimes written as F -Space

♠ **Proposition 3.1.1.2** The following propositions are equivalent:

- (E, P) is Frechet space;
- (E, P) is a LCTVS, and P can be chosen as a countable family.

◦ **Proof**

- $\Rightarrow$
- $\Leftarrow$ Metric

$$d(x, y) := \sum_{j=0}^{\infty} 2^{-j} \frac{P(x, y)}{1 + P(x, y)}$$

□

3.1.2 LF-Space

Definition 3.1.2.1 ...

Completeness

◇ **Lemma 3.1.2.2** Suppose E be LF-Space, with filtration

$$E_0 \subseteq \dots \subseteq E_n \subseteq E_{n+1} \subseteq \dots, E = \cup E_n$$

We have: For sequence $(x_n) \in Seq(E)$,

(x_n) converges $\Leftrightarrow$ There exists some $N \in \mathbb{N}$ such that: $\forall x_n \in E_N, (x_n)$ converges in E_N .

◦ **Proof ...**

□

♠ **Proposition 3.1.2.3** ...

◦ **Proof ...**

□

3.1.3 Hormander's Proposition for $C_c^\infty(\Omega)$

◇ Lemma 3.1.3.1 ...

◦ Proof ...

□

► **Theorem** (Hormander) Suppose $\Omega \subseteq \mathbb{R}^n$. We have

$$C_c^\infty(\Omega) = \lim_{K \subseteq \Omega, K \text{ is compact}} C_c^\infty(K)$$

◦ Proof ...

□

3.2 The Space of Holomorphic Functions in Open Subset of $\mathbb{C}^n$

3.3 The Space of Formal Power Series in n Interminates

3.4 LF Space

Chapter 4 Normable Space, Banach Space

4.1 Norms

4.2 Linear Operators on a Banach Space

4.2.1 The Hille-Yosida Theorem

Chapter 5 Lesbegue Space

Chapter 6 Sobolev Spaces

Chapter 7 Hilbert Spaces

7.1 Basic Definitions

7.1.1 Sesquilinear Form

Definition 7.1.1.1 ...

7.1.2 Pre-Hilbert Space

Definition 7.1.2.1 The pair consisting of a vector space E and a positive definite sesquilinear form B on E is called a **complex Hausdorff pre-Hilbert space**.

7.2 Operators on Hilbert Space

♠ **Proposition 7.2.0.1** (Continuity of Operators) Let $\mathfrak{F}, \mathfrak{K}$ be Hilbert spaces and $A : \mathfrak{F} \rightarrow \mathfrak{K}$ is a linear transformation. The following statements are equivalent:

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-
-
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7.2.1 The Adjoint of An Operator

7.2.2 Projections and Idempotents; Invariant and Reducing Subspaces

7.2.3 Compact Operators

7.2.4 The Diagonalization of Compact Self-Adjoint Operators

7.2.5 Sturm-Liouville Systems

7.2.6 Spectral Theorem and Functional Calculus for Compact Normal Operators

7.2.7 Unitary Equivalence for Compact Normal Operators